In [72]:
```
import pandas as pd

df = pd.read_excel(r'C:\Users\tprac\OneDrive\Documents\customer_shopping_behavior.c
```

In [13]:
```
df.head()
```

Out[13]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise

In [15]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   object 
 3   Item Purchased          3900 non-null   object 
 4   Category                3900 non-null   object 
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   object 
 7   Size                    3900 non-null   object 
 8   Color                   3900 non-null   object 
 9   Season                  3900 non-null   object 
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object 
 12  Shipping Type           3900 non-null   object 
 13  Discount Applied        3900 non-null   object 
 14  Promo Code Used         3900 non-null   object 
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   object 
 17  Frequency of Purchases  3900 non-null   object 
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

In [19]:
```
df.describe(include = 'all')
```

Out[19]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	S
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	39
unique	NaN	NaN	2	25	4	NaN	50	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	
freq	NaN	NaN	2652	171	1737	NaN	96	17
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	N

In [21]:
```
df.isnull().sum()
```

Out[21]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [25]:
```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x
```

In [27]:
```
df.isnull().sum()
```

Out[27]:
```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

In [35]:
```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)' : 'purchase_amount'})
```

In [37]:
```
df.columns
```

Out[37]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

In [39]:
```
# Create a new column age_groups
labels = {'Young Adult', ' Adult', 'Middle-aged', 'Senior'}
df['age_groups'] = pd.qcut(df['age'], q=4, labels = labels)
```

In [41]:
```
df[['age', 'age_groups']]
```

Out[41]:

	age	age_groups
0	55	Young Adult
1	19	Senior
2	50	Young Adult
3	21	Senior
4	45	Young Adult
...	...	...
3895	40	Middle-aged
3896	52	Young Adult
3897	46	Young Adult
3898	44	Middle-aged
3899	52	Young Adult

3900 rows × 2 columns

In [43]:
```
# Create feature columns purchase_frequency_days
df['frequency_of_purchases'].unique()
```

Out[43]:
```
array(['Fortnightly', 'Weekly', 'Annually', 'Quarterly', 'Bi-Weekly',
       'Monthly', 'Every 3 Months'], dtype=object)
```

In [49]:
```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Annually': 365,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Monthly': 30,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

In [51]:
```
df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

Out[51]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

In [53]:
```
df[['discount_applied', 'promo_code_used']].head(10)
```

Out[53]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

In [59]:
```
(df['discount_applied'] == df['promo_code_used']).all()
```

Out[59]:
```
True
```

In [61]:
```
df = df.drop('promo_code_used', axis=1)
```

In [63]:
```
df.columns
```

Out[63]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_groups', 'purchase_frequency_days'],
      dtype='object')
```

In [66]:
```
pip install psycopg2-binary sqlalchemy
```

```
Requirement already satisfied: psycopg2-binary in c:\users\tprac\anaconda3\lib\site-
packages (2.9.13)Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: sqlalchemy in c:\users\tprac\anaconda3\lib\site-packa
ges (2.0.37)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\tprac\anaconda3
\lib\site-packages (from sqlalchemy) (4.15.0)
Requirement already satisfied: greenlet!=0.4.17 in c:\users\tprac\anaconda3\lib\site
-packages (from sqlalchemy) (2.0.1)
```

In [70]:
```
from sqlalchemy import create_engine

username = "postgres"
password = "Wachinga"
host = "localhost"
port = "5432"
database = "customer_behavior"

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/

# Load DataFrame into PostgreSQL
table_name = "customer"
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"(Data successfully loaded into table '{table_name}' in database '{database}
```

```
(Data successfully loaded into table 'customer' in database 'customer_behavior'.
```

In [74]:
```
pip install nbconvert
```

```
Requirement already satisfied: nbconvert in c:\users\tprac\anaconda3\lib\site-packag
es (7.16.4)Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: beautifulsoup4 in c:\users\tprac\anaconda3\lib\site-p
ackages (from nbconvert) (4.12.2)
Requirement already satisfied: bleach!=5.0.0 in c:\users\tprac\anaconda3\lib\site-pa
ckages (from nbconvert) (4.1.0)
Requirement already satisfied: defusedxml in c:\users\tprac\anaconda3\lib\site-packa
ges (from nbconvert) (0.7.1)
Requirement already satisfied: jinja2>=3.0 in c:\users\tprac\anaconda3\lib\site-pack
ages (from nbconvert) (3.1.2)
Requirement already satisfied: jupyter-core>=4.7 in c:\users\tprac\anaconda3\lib\sit
e-packages (from nbconvert) (5.3.0)
Requirement already satisfied: jupyterlab-pygments in c:\users\tprac\anaconda3\lib\s
ite-packages (from nbconvert) (0.1.2)
Requirement already satisfied: markupsafe>=2.0 in c:\users\tprac\anaconda3\lib\site-
packages (from nbconvert) (2.1.1)
Requirement already satisfied: mistune<4,>=2.0.3 in c:\users\tprac\anaconda3\lib\sit
e-packages (from nbconvert) (2.0.4)
Requirement already satisfied: nbclient>=0.5.0 in c:\users\tprac\anaconda3\lib\site-
packages (from nbconvert) (0.5.13)
Requirement already satisfied: nbformat>=5.7 in c:\users\tprac\anaconda3\lib\site-pa
ckages (from nbconvert) (5.9.2)
Requirement already satisfied: packaging in c:\users\tprac\anaconda3\lib\site-packag
es (from nbconvert) (23.1)
Requirement already satisfied: pandocfilters>=1.4.1 in c:\users\tprac\anaconda3\lib
\site-packages (from nbconvert) (1.5.0)
Requirement already satisfied: pygments>=2.4.1 in c:\users\tprac\anaconda3\lib\site-
packages (from nbconvert) (2.15.1)
Requirement already satisfied: tinycss2 in c:\users\tprac\anaconda3\lib\site-package
s (from nbconvert) (1.2.1)
Requirement already satisfied: traitlets>=5.1 in c:\users\tprac\anaconda3\lib\site-p
ackages (from nbconvert) (5.7.1)
Requirement already satisfied: six>=1.9.0 in c:\users\tprac\anaconda3\lib\site-packa
ges (from bleach!=5.0.0->nbconvert) (1.16.0)
Requirement already satisfied: webencodings in c:\users\tprac\anaconda3\lib\site-pac
kages (from bleach!=5.0.0->nbconvert) (0.5.1)
Requirement already satisfied: platformdirs>=2.5 in c:\users\tprac\anaconda3\lib\sit
e-packages (from jupyter-core>=4.7->nbconvert) (3.10.0)
Requirement already satisfied: pywin32>=300 in c:\users\tprac\anaconda3\lib\site-pac
kages (from jupyter-core>=4.7->nbconvert) (305.1)
Requirement already satisfied: jupyter-client>=6.1.5 in c:\users\tprac\anaconda3\lib
\site-packages (from nbclient>=0.5.0->nbconvert) (7.4.9)
Requirement already satisfied: nest-asyncio in c:\users\tprac\anaconda3\lib\site-pac
kages (from nbclient>=0.5.0->nbconvert) (1.5.6)
Requirement already satisfied: fastjsonschema in c:\users\tprac\anaconda3\lib\site-p
ackages (from nbformat>=5.7->nbconvert) (2.16.2)
Requirement already satisfied: jsonschema>=2.6 in c:\users\tprac\anaconda3\lib\site-
packages (from nbformat>=5.7->nbconvert) (4.23.0)
Requirement already satisfied: soupsieve>1.2 in c:\users\tprac\anaconda3\lib\site-pa
ckages (from beautifulsoup4->nbconvert) (2.4)
Requirement already satisfied: attrs>=22.2.0 in c:\users\tprac\anaconda3\lib\site-pa
ckages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (24.3.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in c:\users\tpra
c\anaconda3\lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (202
3.7.1)
```

```
Requirement already satisfied: referencing>=0.28.4 in c:\users\tprac\anaconda3\lib\s
ite-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.30.2)
Requirement already satisfied: rpds-py>=0.7.1 in c:\users\tprac\anaconda3\lib\site-p
ackages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.22.3)
Requirement already satisfied: entrypoints in c:\users\tprac\anaconda3\lib\site-pack
ages (from jupyter-client>=6.1.5->nbclient>=0.5.0->nbconvert) (0.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\tprac\anaconda3\li
b\site-packages (from jupyter-client>=6.1.5->nbclient>=0.5.0->nbconvert) (2.8.2)
Requirement already satisfied: pyzmq>=23.0 in c:\users\tprac\anaconda3\lib\site-pack
ages (from jupyter-client>=6.1.5->nbclient>=0.5.0->nbconvert) (25.1.0)
Requirement already satisfied: tornado>=6.2 in c:\users\tprac\anaconda3\lib\site-pac
kages (from jupyter-client>=6.1.5->nbclient>=0.5.0->nbconvert) (6.3.2)
```

In [78]:
```
pip install pandoc
```

```
Collecting pandocNote: you may need to restart the kernel to use updated packages.

  Downloading pandoc-2.4.tar.gz (34 kB)
  Preparing metadata (setup.py): started
  Preparing metadata (setup.py): finished with status 'done'
Collecting plumbum (from pandoc)
  Obtaining dependency information for plumbum from https://files.pythonhosted.org/p
ackages/b5/55/fdd01894807d20fb6181f13590ae247585b10e909b42e6ada833379fe520/plumbum-
2.0.2-py3-none-any.whl.metadata
  Downloading plumbum-2.0.2-py3-none-any.whl.metadata (8.5 kB)
Requirement already satisfied: ply in c:\users\tprac\anaconda3\lib\site-packages (fr
om pandoc) (3.11)
Requirement already satisfied: typing-extensions in c:\users\tprac\anaconda3\lib\sit
e-packages (from plumbum->pandoc) (4.15.0)
Downloading plumbum-2.0.2-py3-none-any.whl (168 kB)
   ---------------------------------------- 0.0/168.8 kB ? eta -:--:--
   ---------------------------------------- 168.8/168.8 kB 3.4 MB/s eta 0:00:00
Building wheels for collected packages: pandoc
  Building wheel for pandoc (setup.py): started
  Building wheel for pandoc (setup.py): finished with status 'done'
  Created wheel for pandoc: filename=pandoc-2.4-py3-none-any.whl size=34819 sha256=7
117cd1f95fd2207c2db7442c041c1fd00743ef0c879f7867a1b83db12b4b327
  Stored in directory: c:\users\tprac\appdata\local\pip\cache\wheels\4f\d7\32\c6c9b7
b05e852e920fd72174487be3a0f18e633a7adcc303be
Successfully built pandoc
Installing collected packages: plumbum, pandoc
Successfully installed pandoc-2.4 plumbum-2.0.2
```

In [ ]:
```
```

In [ ]:
```
```